NOUS ON MEMORYAGENTBENCH: A PAPER-FAITHFUL EVALUATION OF A COGNITIVE MEMORY AGENT

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ABSTRACT

We evaluate nous, a cognitive memory agent with episodic ingestion, LLM-based fact extraction, and sleep-cycle consolidation, on MemoryAgentBench (Hu et al., ICLR 2026), a benchmark that measures four memory competencies of conversational agents: Accurate Retrieval (AR), Test-Time Learning (TTL), Long-Range Understanding (LRU), and Conflict Resolution (CR). We reproduce the benchmark's evaluation protocol exactly—its per-dataset answer prompts, its per-dataset graders (including two GPT-4o judge protocols), vendored from the benchmark's open-source repository and verified line-by-line by an independent review—and drive a real nous HTTP server through its full production cognitive cycle for every test instance. On the evaluated subsets, nous scores **AR 0.836** (n=568, 95% CI [0.81, 0.87]), **CR 0.725** (n=320, CI [0.68, 0.77]; single-hop 0.887, multi-hop 0.562), **LRU 0.824** (n=68, CI [0.73, 0.91]), and **TTL 0.555** (n=200, CI [0.49, 0.62]). The first three are robustly above every method in the benchmark paper's results (best per-competency averages there: 71.8%, 29.5%, 62.2%); TTL sits at the top of that range (53.9%). Against the 2026 field at sub-dataset level — with every already-ingested context now at *full* question coverage, under a reconstruction of its own gpt-5-judge protocol, and with margins graded against known backbone swings ($\sim$9.7pp) — nous and the leading 2026 system are statistical peers overall: one decidable nous win (multi-hop conflict resolution, $+$21.2pp), one decidable loss (in-context test-time learning, $-$21.5pp), one weak cell (LongMemEval, $-$7.6pp), and everything else (event retrieval $+$8.9, single-hop document QA $+$8.0, detective QA $+$8.1, single-hop conflict $+$8.4, multi-hop document QA $+$2.0) within backbone-swing range. We describe the testing technique, a statistical-validation pass that materially corrected one competency's initial estimate, and the threats to validity that qualify a direct leaderboard comparison. A post-baseline causal improvement program (fifteen intervention arms on the same persisted memory) then isolated and fixed three write-path store defects upstream in nous, added a bounded two-round exact-key retrieval leg, and added an exemplar-granularity retrieval mode; the resulting configurations score **CR 0.812** (single-hop 0.938, multi-hop 0.688; p=0.0008 vs. baseline) and **ICL 0.695** (vs. 0.555, p=0.0018, narrowing the program's sole decidable field loss from $-$28.5pp to $-$14.5pp) — each step gated by zero-cost store simulations before any paid evaluation.

1 Introduction

Long-running LLM agents need memory that outlives a single context window: they must store what they were told, learn new skills from it, reason over inputs far larger than the model context, and revise beliefs when information changes. MemoryAgentBench [1] operationalizes these requirements as four competencies and evaluates them by streaming large corpora into an agent incrementally and then asking questions against what the agent retained.

nous is a cognitive AI agent whose memory subsystem combines an episodic store (conversation episodes and summaries), a semantic store (LLM-extracted facts with typed graph edges), full-fidelity content chunks with hybrid (vector + lexical) retrieval, and an offline *sleep* consolidation cycle that links, deduplicates, and densifies memory between sessions. This report documents a from-scratch evaluation harness that measures nous on MemoryAgentBench under its *production* configuration, and places the results against the methods evaluated by the benchmark's authors.

Two principles governed the effort:

1. **Paper fidelity.** The score must be computed exactly as the benchmark computes it—same answer prompts, same graders, same judge models—so numbers are comparable. All grading code was vendored from the benchmark's MIT-licensed repository and cross-reviewed against it by an independent reviewer agent with no stake in the outcome.
2. **Statistical honesty.** No above/below-field claim is made unless the 95% confidence interval of the measured accuracy clears the best published field value. This rule changed one headline conclusion (Section 4.3).

2 System Under Test

2.1 The nous cognitive cycle

This subsection describes the system as it actually runs — module and function names refer to the nous codebase at the evaluated revision, and every mechanism named here was verified against source by an independent code-mapping pass. Each benchmark instance exercises the complete production path:

1. **Ingestion.** The instance's context is split into chunks of at most 32,000 characters (conversational corpora are packed turn-by-turn into chunks of the same size, preserving every turn) and streamed to the agent as consecutive chat turns in one session (`POST /chat`).
2. **Session close.** The session is closed, triggering the durable-write chain described below.
3. **Sleep consolidation.** One or more sleep cycles run the offline consolidation phases described below.
4. **Answering.** Every question is asked in a *fresh* session. The agent answers from its memory (pre-turn retrieval-augmented context injection plus agentic recall tools); the question is framed with the benchmark's own per-dataset prompt template.

Turn orchestration and pre-turn injection. A turn is driven by a cognitive layer phased as SENSE $\rightarrow$ FRAME $\rightarrow$ RECALL $\rightarrow$ DELIBERATE before the model is called, and ACT $\rightarrow$ MONITOR $\rightarrow$ LEARN after (`cognitive/layer.py`). The RECALL phase assembles the system prompt from prioritized sections (`cognitive/context.py::ContextEngine.build`): identity, an anti-hallucination notice, the user profile (tier-1 fact categories), active censors, the selected frame, session working memory, related decisions, relevant facts, recent conversations, and matching past episodes. The per-type fact pipeline applies, in order: recency resolution, staleness penalty, frame boost, subject-diversity enforcement, deduplication, usage boost, a relevance floor, and pinned-fact reinsertion. Section budgets are frame-dependent and environment-overridable; sections are grouped into static/semi-stable/dynamic tiers to exploit prompt caching.

Retrieval. Pre-turn injection uses per-type searches; the deeper pipeline (`api/retrieval_pipeline.py::run_recall_pipeline`) serves the agentic recall tools. Its stages, in execution order, each behind its own flag: (1) *heart recall* over episodes/facts/procedures — internally a reciprocal-rank-fusion hybrid of vector and keyword legs (w_v=0.7, k=60), with optional cross-encoder re-ranking and MMR diversity both *off* in the evaluated configuration (the CE was measured retrieval-negative in this program; it remains in use only for sleep-time edge auditing); (1.5) the *episode-chunk leg* — hybrid vector+FTS search over verbatim transcript chunks, the channel our probes identified as carrying $\sim$65% of answerable content; (1.6) the *keyed fact leg* (F085) — exact entity-key lookup against a bidirectional key index, merged in a bounded score band (K=8); (R3v2) an optional *second keyed round* that feeds round-one hits' entity keys back as new lookup keys under fan-out guards (32 keys / 256 candidates) with a deterministic attribute-overlap ranking (K_2=8); (1.7) the *exemplar leg* (F086) — cosine k-NN over individually-embedded labeled examples, trigger-gated to classification-shaped queries; (2) cross-type graph expansion from heart seeds; (3) decision search; (4) graph expansion via one-hop neighbors or, when a density gate opens, spreading activation (measured negative for dense-fact QA in this program and kept gated off); (5) contradiction detection over `contradicts` edges among the results. Results merge into score bands (direct hits above keyed round-one above keyed round-two), deduplicate across legs, and exclude anything already injected into the current turn's system prompt — so agentic recall deliberately surfaces the next-best alternatives beyond pre-turn injection. The answering agent can invoke `recall_deep` (the full pipeline), `recall_recent` (time-ordered episodes), `get_procedure`, and a graph-hub tool mid-turn.

Write path. Session close ends the episode (trivial single-turn episodes are discarded), persists the transcript, and triggers an asynchronous episode summarizer that also chunks and embeds the transcript verbatim (600-character chunks, 80 overlap). Fact extraction then routes *modally* by content shape: example-shaped streams (`utterance`→`label` density ≥0.8) are parsed — not LLM-extracted — into individually-embedded exemplar facts (F086); enumerable dense-document episodes (statement-line density ≥0.6) route to the enumerative extractor (F084), which emits one fact per statement with subject/attribute keys and per-statement source ordinals; everything else takes the legacy summarizer/extractor path. Every fact passes an admission controller scoring utility, grounding, novelty, recency, and a category prior (threshold 0.55; corrective sources such as user-stated facts and the extraction legs bypass scoring). Near-duplicates are detected at cosine ≥0.95 — with the critical exception, added by this program's Gate-1 fix, that a same-(`subject_key, attribute_key`) pair with a *different value* routes to conflict resolution instead of being confirmed away as a duplicate. Conflict resolution is resolved by *statement order* (same-episode source ordinal, else `learned_at`); the LLM conflict classifier is advisory relation-typing only, and pairs with no ordering signal are kept both and flagged with a `contradicts` edge. On insert, each fact's subject and value-side entities are written to a canonical-normalized entity-key index (at most 8 keys/fact) that serves the keyed retrieval legs.

Sleep consolidation. An idle-triggered background cycle runs phases ordered free-first/LLM-last: decision-outcome review, pruning, episode compression, cross-session reflection, embedding-based contradiction resolution, the key-conflict supersession sweep, stale-fact retirement, same-subject cluster consolidation, abandoned-episode recovery, graph densification (where sleep-time cross-encoder edge auditing applies), episode-edge relinking, dead-edge pruning, and procedure generalization. Background phases use a separate background model from the answering model.

Storage. Postgres with `pgvector`: `heart.facts` (content, category, confidence, provenance, `subject_key/attribute_key`, `source_ordinal`, `superseded_by/active` lifecycle, 1536-dim embedding, generated FTS column); `heart.fact_entity_keys` (the bidirectional key index, btree on `(agent_id, entity_key)`); `heart.episode_chunks` (verbatim transcript chunks with HNSW vector and GIN FTS indexes); `heart.episodes`; and `brain.graph_edges` (typed relations including `supersedes/contradicts/co_occurred`, with extraction-method provenance). Exemplar facts get a dedicated *partial* HNSW index scoped to active exemplar rows so the k-NN walk never touches the general embedding index.

2.2 Configuration

nous ran under its live production configuration — Claude Sonnet 5 as the reasoning model (`NOUS_MODEL` overriding the code default), Sonnet-class background model for extraction and consolidation, a Haiku-class model for the advisory conflict classifier, `text-embedding-3-large` at 1536 dimensions for all embeddings, hybrid chunk search enabled, date-aware retrieval, adaptive thinking — with exactly three eval-hygiene overrides, all disabling background self-improvement features irrelevant to memory measurement: the heartbeat timer, the scheduler, and rubric outcome detection. Retrieval-time cross-encoder re-ranking, MMR, and spreading activation were off (the first and third measured negative in this program; Section 5.4). A 117-knob snapshot of the configuration is pinned in the harness repository for reproducibility.

Two run profiles were used. The *full-fidelity* profile (unbounded episode summarization, three sleep cycles) is used for moderate contexts. For multi-megabyte contexts a *big-context* profile bounds the episode summarizer to 8 chunks, runs one sleep cycle, and paces ingestion turns (15 s); this profile eliminated upstream rate-limit stalls and ingest timeouts entirely (0 errors across all final runs, including 2.3M-character documents).

2.3 Isolation

Every instance gets a fresh nous server process, a unique agent identity, and a dedicated evaluation database; nothing carries over between instances. Control probes confirmed the isolation: on Conflict Resolution, the identical questions asked of an agent *before* ingestion score 0.000, so the measured accuracy is attributable to memory, not to model priors.

3 Benchmark and Datasets

MemoryAgentBench streams each dataset's context into the agent and then asks held-out questions. Table 1 lists the evaluated subsets.

	Source	Context size	Questions	Grader
CR	factconsolidation_sh (6k–262k)	26k–1.1M	4×40	substring (SQuAD-norm.)
	factconsolidation_mh (6k–262k)	26k–1.1M	4×40	substring (SQuAD-norm.)
	single-hop (sh) = updated fact; multi-hop (mh) = compose two updated facts			
AR	eventqa_65536	∼285k	200 (5 ctx×40)	all-elements recall
	eventqa_131072	∼570k	100 (1 ctx)	all-elements recall
	eventqa_full	∼2.3M	8	all-elements recall
	SH-/MH-Doc QA (ruler_qa1/qa2)	0.9M / 1.9M	100+100	substring
	longmemeval_s* (1282 turns)	1.6M	60 (full, 1 of 5 ctx)	GPT-4o yes/no judge
TTL	icl_banking77 / clinic150 / nlu	∼400k ea.	3×40	parsed-label exact
	icl_trec_coarse / trec_fine	∼400k ea.	2×40	parsed-label exact
LRU	detective_qa (10 novels)	∼400k ea.	68	substring
	infbench_sum_eng	1.75M	1	GPT-4o 3-call f1 judge

Table 1: Evaluated datasets. "Context size" is the raw document streamed into memory. The TTL *recsys* sub-dataset could not be graded faithfully: its grader requires an entity-id mapping file that is not shipped in the benchmark's public repository, so TTL here is the in-context-learning family only.

4 Testing Technique

4.1 Paper-faithful grading

Each sub-dataset is graded exactly as in the benchmark's `post_process` dispatch:

- **Substring match** (ruler, detective, factconsolidation): the SQuAD-style normalization (lowercase, strip punctuation and articles, collapse whitespace), then gold-in-prediction containment, max over golds.
- **EventQA recall**: *raw* case-insensitive containment of every gold element (`el.lower() in pred.lower()`); the answer is correct only if all elements are present. An initial implementation that reused the SQuAD normalization here was flagged by independent review as strictly more lenient than the paper (punctuation and article stripping create false positives) and was corrected before any reported run.
- **ICL exact match**: parse the label the prompt requests (`label: X`), then exact match on the bare label. This is a deliberate, conservative interpretation: the benchmark's own ICL scoring parses with a different prefix and reports multiple metrics of which the substring variant false-positives on digit collisions (gold `5` matches `label: 15`); our parser cannot.
- **LongMemEval judge**: the benchmark's GPT-4o yes/no judge with its verbatim per-question-type prompts (including the temporal-reasoning off-by-one tolerance and the abstention protocol for `_abs` questions).
- **Summarization judge** (InfBench-Sum): the benchmark's three-call GPT-4o protocol scoring fluency $f \in \{0, 1\}$, key-point recall r, and sentence-level precision p, combined as $\mathrm{f1} = f \cdot \frac{2rp}{r+p}$.

Answer prompts are likewise the benchmark's own per-dataset templates (e.g., the CR template that states the serial-number update rule with few-shot examples; the eventQA template that frames "what happened next"). On identical memory, switching from a generic answer prompt to the benchmark's CR template moved CR accuracy by +28 percentage points—prompt fidelity is not optional if numbers are to be compared.

4.2 Independent review

All vendored grading and prompting code was audited by a reviewer agent with fresh context against the benchmark's repository, file by file. The review confirmed byte-fidelity of the normalizers, judge prompts, dispatch tables, and the summarization formula; it found exactly one fidelity bug (the eventQA normalization above), which was fixed and regression-tested before the reported runs. The harness carries 125 unit tests covering graders, judges, chunking, persistence, and isolation.

4.3 Statistical validation

Initial runs used 8 questions per source. A dedicated validation pass then grew the sample until each competency's 95% CI either cleared the best published field value or demonstrably did not:

- **TTL corrected downward.** At n=8/source TTL measured 0.650; at n=40/source (n=200) it fell to **0.555** (CI [0.486, 0.624]), with one source falling from 0.625 to 0.450. The small sample had been an optimistic draw. TTL is therefore reported as *at* the field ceiling, not above it.
- **LRU confirmed.** Detective-QA at all 10 novel contexts (n=68) held at **0.824** versus 0.833 at n=6; its lower CI bound (0.733) clears the best field value (0.622).
- **AR firmed upward.** EventQA-65536 across 5 contexts at n=200 rose to 0.910 versus 0.875 at n=24, consolidating AR at **0.897** (CI [0.857, 0.936]).
- **CR corrected downward — via replay, not re-ingest.** At n=8/instance CR measured 0.766; replaying *all 40 questions per instance against the same persisted memory* (n=320; a pure sample-size extension with the memory-build variable held fixed, content-verified agent→instance mapping) re-priced it to **0.725** (CI [0.676, 0.774]): single-hop 0.887, multi-hop 0.562, with multi-hop degrading monotonically with context size (0.825/0.600/0.525/0.300 at 6k/32k/64k/262k).
- **AR corrected by full coverage — replay again.** Extending every already-ingested AR context to its complete question set (n=568 total) moved every thin cell: LongMemEval fell 0.875 (n=8) → **0.700** (n=60, including the first measurement of its abstention questions), MH-Doc QA fell 0.750 → **0.710** (n=100), while eventqa-131072 *rose* 0.750 → **0.860** (n=100) and SH-Doc QA held (0.875 → 0.870, n=100). Consolidated AR: 0.897 (n=232) → **0.836** (n=568, CI [0.806, 0.867]) — still robustly above the benchmark paper's field, no longer clearly above the 2026 leader.

The general lesson—small-sample memory-eval estimates run optimistic; grow n until the CI clears the comparison band—is recorded as a standing methodological rule for this project. All results persist per-question to append-only JSONL as each instance completes, so partial runs are never lost and every reported number is recomputed from the persisted records.

5 Results

5.1 nous scores

Competency	Score	n	95% CI	Best published avg. / verdict
Accurate Retrieval	**0.836**	568	[0.806, 0.867]	0.718 — **above, robust**
Conflict Resolution	**0.725**	320	[0.676, 0.774]	0.295 — **above, robust**
Long-Range Understanding	**0.824**	68	[0.733, 0.914]	0.622 — **above, robust**
Test-Time Learning (ICL subset)	**0.555**	200	[0.486, 0.624]	0.539 — at ceiling

Table 2: nous on MemoryAgentBench (paper-faithful grading, production configuration, 0 execution errors on all final runs). LRU is detective-QA; the auxiliary InfBench-Sum summarization judge scored f1 = 0.592 on a single 1.75M-char book (unfirmed, n=1). "Best published avg." is the strongest per-competency average across all methods in the benchmark's results table. *Comparability tiers:* AR, CR, and LRU are **T1** (paper's exact prompt/grader/questions; sole deviation is ingestion transport, Section 6); TTL is **T2** (T1 plus a deliberate, conservative label-parsing choice in the ICL grader). No single aggregate score is published: TTL omits recsys (grader data file not publicly shipped), and the giant AR/LRU sources are measured at 1 context each. The repository's `BASELINE_SUMMARY.md` reconciles every number here to its committed per-question evidence file, tier, and missing-components list.

Per-source detail: CR (n=320: the full 40 questions per instance, replayed against the same persisted memory that produced the initial run) single-hop 0.887 vs. multi-hop 0.562 (the benchmark paper reports at most ∼7% for multi-hop across its methods); multi-hop degrades with context size: 0.825/0.600/0.525/0.300 at 6k/32k/64k/262k. AR eventqa 0.910/0.860/0.875 (65k n=200 / 131k n=100 / full n=8), SH-Doc/MH-Doc QA 0.870/0.710 (n=100 each; MAB's official names — our source ids ruler_qa1_197K/qa2_421K), longmemeval 0.700 (n=60); TTL banking77 0.525, clinic150 0.500, nlu 0.425, trec-coarse 0.875, trec-fine 0.450; LRU detective 56/68.

5.2 Comparison with the benchmark paper's field (2025)

Table 3 reproduces the per-competency averages of representative methods from the benchmark's results table (its Table 3; the fourth competency is named Selective Forgetting there and Conflict Resolution in earlier versions—same datasets). These rows date from the benchmark paper; the 2026 field is covered in Section 5.3.

Three observations against the paper's table:

Method	Class	AR	TTL	LRU	CR/SF
GPT-4o-mini (full context)	long-context	49.2	48.6	46.2	25.0
GPT-4.1-mini (full context)	long-context	71.8	46.2	49.1	20.5
Claude-3.7-Sonnet (full context)	long-context	59.7	**53.9**	**62.2**	22.5
BM25	RAG	60.5	44.5	35.6	25.5
Text-Embed-3-Large	RAG	54.6	44.3	37.3	16.0
HippoRAG-v2	RAG	65.1	35.8	36.2	**29.5**
Mem0	agentic memory	32.6	21.2	20.7	10.0
GraphRAG	structure-aug.	40.9	24.8	19.9	8.0
MIRIX	agentic memory	47.5	24.1	25.4	8.0
nous (this work)	cognitive memory	**83.6**	55.5[§]	**82.4**	**72.5**

Table 3: Per-competency averages (%). Field rows are from the benchmark's published results; bold marks the best value per column among field methods, and nous where it exceeds all of them. [§] ICL subset only (recsys excluded). Comparison caveats in Section 6 apply—in particular the base-model mismatch and the subset sizes.

1. **Conflict Resolution is the standout vs. the 2025 field.** nous scores 0.725 where no method in the paper exceeds 0.295, and 0.562 on multi-hop conflict composition where that field is at most ∼0.07. We attribute this to nous's explicit fact store with update semantics: conflicting facts are extracted, linked, and consolidated during sleep, rather than left as contradictory passages for retrieval to stumble over.
2. **Retrieval and long-range understanding benefit from full-fidelity chunk memory.** AR 0.836 and LRU 0.824 exceed both the RAG family and the long-context family, including on 1.6–2.3M-char inputs that no context window holds.
3. **Test-time learning is the honest gap.** TTL 0.555 edges the best value in the paper (0.539) but the CI straddles it. Learning a classification skill from thousands of in-context examples is not what an episodic/semantic memory is optimized for; accuracy here tracks how many labeled exemplars retrieval can surface per query.

5.3 Comparison with the 2026 field — sub-dataset level, dual-graded

The field has moved since the benchmark paper. Infini Memory [2] (2026) re-evaluated a modern method suite on the *full* benchmark with gpt-5-mini as the base model, 4096-token ingestion chunks, and a gpt-5 LLM-as-judge protocol (binary correctness from question, reference, and output; no judge prompt is published). Every row of its Table 1 was verbatim-extracted and arithmetically verified (each group average recomputes from its sub-values; each overall from the four group averages).

To make the comparison protocol-matched in both directions, we additionally **re-graded every persisted nous answer with a reconstruction of their judge protocol** (gpt-5, binary, question + reference + output) — pure post-processing over the committed evidence files. Table 4 reports the field at sub-dataset level with nous under both gradings.

Backbone-adjusted read. Every nous-vs-Infini comparison below carries the backbone confound (claude-sonnet-5 vs. gpt-5-mini)[‡]. The benchmark's own ablations show backbone swings of order ∼9.7pp for agentic-memory methods (MIRIX, GPT-4o-mini→GPT-4.1-mini), so we grade each margin against that yardstick rather than treating any lead as architecture-attributable:

- **FC-MH (multi-hop conflict, n=160): margin +21.2pp (56.2 vs. 35.0) — exceeds the known backbone swing**; the one comparative claim that survives the confound, robust under all three grading protocols (0.562 official, 0.562 gpt-5, 0.469 gpt-4o — all above 0.350). It is also the cell Infini's authors identify as unsolved ("multi-hop forgetting cannot be settled at write time").
- **ICL: −21.5pp behind (62.5 vs. 84.0)** — the one decidable loss; a real capability gap, not a grading artifact, plausibly reflecting Infini's up-to-seven-round agentic retrieval loop surfacing more labeled exemplars per query.
- **AR: margin +2.8pp (84.0 vs. 81.2) — statistical peer.** Full coverage collapsed the earlier +10.6pp margin: the thin n=8 cells (LME, MH-Doc QA) had been optimistic draws. Within AR, Event (+8.9) and SH-Doc QA (+8.0) lead, MH-Doc QA is a wash (+2.0), and **LME is nominally behind (−7.6, 71.7 vs. 79.3; Wilson 95% CI on our n=60 cell [0.58, 0.80])** — with a construct-validity caveat: Infini's table labels LME a "reconstructed multi-session variant," which matches MemoryAgentBench's own `longmemeval_s*` naming

	Accurate Retrieval					TTL			LRU			CR/SF			
Method	SH	MH	LME	Evt	*Avg*	ICL	Rec	*Avg*	Sum	Det	*Avg*	SH	MH	*Avg*	Ovr.
Mem0	28.0	36.0	42.0	35.5	35.4	35.2	11.2	23.2	12.4	42.3	27.4	25.0	3.0	14.0	25.0
RAPTOR	32.0	39.0	38.2	48.2	39.4	56.8	14.0	35.4	25.3	45.6	35.5	16.0	2.0	9.0	29.8
MemGPT	49.0	30.0	42.7	45.6	41.8	65.6	14.3	40.0	10.9	40.2	25.6	32.0	3.0	17.5	31.2
MemoRAG	35.0	36.0	25.0	54.4	37.6	75.0	15.7	45.4	15.3	53.5	34.4	25.0	8.0	16.5	33.5
LightMem	57.0	34.0	69.2	39.2	49.9	58.0	12.7	35.4	27.5	41.0	34.2	29.0	6.0	17.5	34.2
REMem	62.0	42.0	62.3	41.2	51.9	80.0	13.2	46.6	23.4	47.6	35.5	32.0	9.0	20.5	38.6
HippoRAG-v2	78.0	70.0	54.7	72.2	68.7	65.2	12.3	38.8	32.1	54.3	43.2	58.0	5.0	31.5	45.5
Infini Mem.-H	77.0	76.0	76.0	83.0	78.0	81.0	15.8	48.4	57.2	78.4	67.8	80.0	22.0	51.0	61.3
Infini Mem.-A	83.0	79.0	79.3	83.6	81.2	**84.0**	18.0	51.0	59.9	77.2	68.6	81.0	35.0	58.0	64.7
nous (official)	87.0	71.0	70.0	89.3	79.3	55.5	—	n/a	59.2	82.4	70.8	88.7	56.2	72.5	n/a
nous (gpt-5 judge)	**91.0**	**81.0**	71.7	**92.5**	**84.0**	62.5	—	n/a	59.2	**85.3**	**72.3**	**89.4**	**56.2**	**72.8**	n/a

Table 4: 2026 field (Infini Memory [2] Table 1, full benchmark, gpt-5-mini base, gpt-5 judge; verbatim-verified) vs. nous slices under both gradings. nous sample sizes: SH-QA/MH-QA n=100 each (full question sets), LME n=60 (full set of its context, incl. abstention questions), Event pooled n=308, ICL n=200, Summ n=1, DetQA n=68, FC-SH/FC-MH n=160 each — all replayed on the same persisted memory that produced the original runs. Rec is data-blocked, so nous TTL Avg and Overall are honestly n/a. nous's judge is a disclosed reconstruction (theirs is unpublished). Base models differ (claude-sonnet-5 vs. gpt-5-mini).

but is not confirmed to be the identical question set, and our cell covers 1 of 5 LME contexts. We therefore report the gap as indicative, not a decided weak cell.

- **FC-SH (+8.4pp) and DetQA (+8.1pp): within the backbone-swing range — directionally ahead but backbone-undecidable**; we do not claim these as architecture wins.

Averaging note: the AR "Avg" column follows Infini's method (macro mean of four sub-dataset accuracies) for apples-to-apples — a method that is statistically weak on both sides (it weights a 60-item cell equally with a 308-item cell). nous's item-weighted (micro) AR is 0.836 official / 0.880 judge-graded, vs. macro 79.3 / 84.0. Two secondary findings: the strict-vs-judge delta is +0–12.5pp on verbose-answer tasks and ≈0 on short-entity tasks (CR), so official grading was a conservative floor; and nous multi-hop conflict resolution **degrades with context scale** (0.825/0.600/0.525/0.300 at 6k/32k/64k/262k). Whether that degradation is retrieval completeness, cross-document supersession propagation, or intrinsic multi-hop reasoning difficulty (the benchmark's own ablation shows o4-mini falling 80%→14% on FC-MH as context grows) requires per-failure triage of the retrieved context; we report the curve as a finding, not a diagnosis.

5.4 Post-baseline: a causal improvement program closes at 0.812

After the baseline froze, we ran a twelve-arm intervention program on the CR competency using the replay-without-reingest technique (Section 4): every arm re-answers the same 320 questions against the same persisted memory, so arms are directly comparable and cost no re-ingestion. Eight read-path and behavior arms (retrieval budget, fact pinning, supersession-lineage formatting, recall backstop, graph anchors, spreading activation ×3, cross-encoder reranking, forced deterministic recall) all measured null-to-negative, clustering in 0.675–0.738. The program then moved to the write path, where offline probes located three store-level defects in sequence: extraction summarized instead of enumerating (the fact store held answerable golds for 1% of questions); cosine deduplication silently dropped stated updates whose embedding matched the fact they were meant to supersede (∼39% of single-hop answer statements never became facts); and the supersession classifier adjudicated conflicts by world knowledge, so a stated update that contradicts the model's prior lost to it (21/160 single-hop golds deactivated) — the precise failure mode a conflict-resolution memory exists to prevent. All three were fixed upstream in nous (enumerative extraction; same-slot routing past dedup; statement-order winner selection with the classifier demoted to advisory relation-typing), each fix gated by zero-cost store simulations before any paid evaluation: post-repair, exact entity-key retrieval of the gold fact reached 0.85 on single-hop (bar 0.80; pre-fix 0.48) with median candidate sets of 1.5 facts.

The decisive arm — the baseline configuration plus a single flag enabling a bounded exact entity-key retrieval leg (K=8, merged below the direct-hit head) over the repaired store — scored **0.759** (243/320; single-hop 0.900, multi-hop 0.619), against 0.725 for the published baseline. Two qualifications keep this honest: the question-level paired sign test is not individually significant (+36/−25, p=0.20), and the claim therefore rests on the arm-distribution comparison — **0.759 exceeds all twelve prior arms** (range 0.675–0.738) — plus the mechanically verified causal chain (store repaired → retrieval gate passed → arm improved). Notably, the gain was multi-hop-driven (+9 net vs. +2 single-hop), contradicting our pre-registered single-hop-driven forecast: the best available explanation is that the agent's answer-time recall is already iterative, and repairing the store improves every retrieval round it performs, not only the new keyed leg — consistent with an offline composition simulation in which feeding round-one fact entities back as round-two keys lifts multi-hop gold coverage from 0.02 to 0.49 on the repaired store (versus approximately nothing pre-repair).

That simulation became the next shipped feature. nous implemented the iterative round as a bounded, deterministic second keyed fetch (keys derived from round-one hits' own entity index rows, fan-out capped, candidates ranked by attribute-word overlap with the query, top-K_2=8 merged in a score band below round one), with the ranking function carrying an explicit sim-parity contract: any change to it requires re-simulation before paid evaluation. The acceptance gate ran against the implementation's own functions (not a re-simulation of the spec): multi-hop keyed-composition coverage 0.42 at K_2=8 against a pre-registered bar of 0.35. The decisive arm — identical to the 0.759 configuration except `rounds`=2 — scored **0.812** (260/320; single-hop 0.938, multi-hop 0.688; CI [0.770, 0.855]; 0 errored). This is the program's first *question-level-significant* improvement: paired against the published baseline, +47/−19 (sign test p=0.0008); paired against the rounds=1 arm, +39/−22 (p=0.040); and the CI lower bound clears the baseline, the twelve-arm noise band, and the rounds=1 point estimate. Multi-hop at 6k reached 0.900, the highest multi-hop cell recorded in this program. Across the program, CR moved 0.725 → 0.759 → 0.812 (+8.7pp), each step mechanically explained and simulation-gated before spend. Two calibration notes carry forward: our pre-registered forecasts under-shot both decisive arms in the same direction — single-hop gained even when the simulations attributed it nothing — because static coverage simulations lower-bound live effects when the agent's own recall loop iterates over the improved substrate. The multi-hop × scale cell remains the frontier (0.475 at 262k, the least-improved cell, exactly as its 0.23 gate coverage predicted).

The program then turned to its sole decidable loss, in-context test-time learning. An independent methodology review first corrected our own accounting — the live ICL baseline file had concatenated two runs, double-counting 40 questions; the true deduplicated baseline is **0.555**, not 0.571 — and refuted a first-pass oracle-style ceiling claim (presence of the gold label in a gathered set is not identifiability). The corrected diagnosis: storage is lossless (0.2–2% chunk-boundary loss) and the model is capable; the 26pp gap to the leader is *retrieval granularity* — stored chunks bury ∼40 labeled exemplars each, so chunk-level search returns a similar region rather than the k nearest labeled examples. A zero-LLM simulation at exemplar granularity (individually-embedded `utterance→label` pairs) showed a deterministic majority-vote-of-5 rule reaching **0.82** — the 2026 leader's level — on nous's own stored memory (paired vs. live, p=8×10^{-9}). nous shipped the corresponding feature (parse-only exemplar extraction plus a bounded, land-dark retrieval leg); the implementation gate reproduced 0.80 against the shipped index; and the decisive n=200 replay scored **0.695** (+52/−24 paired vs. 0.555, p=0.0018) — the largest single-competency gain of the program, narrowing the field loss from −28.5pp to −14.5pp, but the first arm to land *below* its gate. The conversion leak is precisely attributable: where the agent's answer followed the injected exemplar majority (136/200) it scored 0.83, matching the gate; where it overrode (64/200) it scored 0.41 while the ignored majority was right ∼73% of the time. The residual is therefore answer-time reader discretion, not retrieval — the mirror image of the substrate arms, which converted *above* their simulations. Both calibration asymmetries are now part of the program's forecasting method.

6 Threats to Validity

1. **Base-model mismatch.** nous reasons with Claude Sonnet 5; the benchmark's agentic-memory rows mostly use GPT-4o-mini, and its strongest long-context rows use GPT-4.1-mini and Claude-3.7-Sonnet. Part of nous's margin is attributable to the stronger base model. The long-context Claude-3.7 row (best field TTL and LRU) is the fairest anchor, and nous still exceeds it on AR (+30pp), LRU (+20pp), and CR (+54pp).
2. **Subsets, not the full benchmark.** Question counts per source range from 8 to 200 (Table 1); the giants (eventqa-full, MH-Doc QA 421K, longmemeval) were measured at one context each. CIs in Table 2 reflect this. recsys (TTL) is excluded for lack of the grader's data file; InfBench-Sum is a single book.
3. **Configuration provenance.** The production configuration was improved using failures observed on this benchmark's CR subset in an earlier phase (write-path fidelity, hybrid chunk retrieval, recall depth). The evaluation is therefore best read as "nous after its memory bugs were fixed," not as a blind first attempt. No knob was tuned per-dataset, and the same configuration serves all four competencies.
4. **Ingestion transport differs from the paper's protocol.** The benchmark feeds context in chunks of 512 tokens for the doc-QA, LongMemEval, and conflict-resolution tasks and 4096 tokens elsewhere (uniformly 4096 for the commercial memory agents, for cost); our harness streams 32,000-character turns, and nous re-chunks internally for retrieval, so transport granularity is not the retrieval granularity. The baseline is therefore paper-faithful in prompts, graders, judges, and question sets, but *not* an exact reproduction of the paper's ingestion transport. No answer-bearing text was lost to this difference: ingest capacity was 80 × 32k chars (2.56M), and a post-hoc audit of all 34 evaluated instances at the exact run settings confirmed **zero**

dropped chunks (largest instance: 2.31M chars, 73/80 chunks; the 1,282-turn LongMemEval conversation packed to 50 chunks with every turn retained). Runs now stamp `chunks_sent/chunks_truncated` on every persisted result row, and truncated instances are excluded from headline aggregates by construction.

5. **Benchmark prompts embed task rules.** The benchmark's own CR and eventQA templates tell the agent the update rule / task framing. We use them because the field numbers use them; they help any system.
6. **Judge sharing.** LongMemEval and InfBench-Sum scores depend on GPT-4o judges, as in the benchmark.
7. **The 2026 comparison is as-reported, not re-run.** Table 4 takes the 2026 numbers from the Infini Memory paper [2] at face value; we did not reproduce them. Their protocol differs from the benchmark's official grading (gpt-5 LLM-judge for all tasks, typically more lenient than substring graders; 4096-token chunks) and from ours, and their coverage is the full benchmark where ours is slices. We mitigated the grading asymmetry by re-grading all persisted nous answers under a reconstruction of their judge protocol (Table 4), but their exact judge prompt is unpublished and our base model differs, so the comparison is indicative, not exact; we make no state-of-the-art claim.

7 Conclusion

On the sampled, paper-aligned slices and under the benchmark's exact grading, nous exceeds every method reported in the MemoryAgentBench paper on three of four memory competencies—Accurate Retrieval (0.836), Conflict Resolution (0.725 at n=320, a 2.5× margin over that table's best), and Long-Range Understanding (0.824)—and matches that table's best Test-Time Learning (0.555, ICL subset). Against the 2026 field at sub-dataset level, at full question coverage of every ingested context, under a reconstruction of its own judge protocol, and with every margin graded against the ∼9.7pp backbone-swing yardstick (Section 5.3), the honest placement is: **nous and the 2026 leader are statistical peers** — one decidable nous win (**multi-hop conflict resolution**, +21.2pp, tri-protocol robust, on the cell the leader's authors themselves call unsolved), one decidable loss (**ICL test-time learning**, −21.5pp), one nominally-behind cell (**LongMemEval**, −7.6pp, comparability unconfirmed), and every other margin (+2.0 to +8.9pp) within backbone-swing range. One dose-response architectural finding stands: **multi-hop conflict resolution degrades monotonically with context size** (0.825/0.600/0.525/0.300 at 6k/32k/64k/262k — length is the x-axis, n=40 per point). The MH-Doc QA and LME corrections, by contrast, are sampling corrections (their x-axis was n, not length; each queries a single fixed-length context), and we report them as such: our n=8 estimates were optimistic, full coverage fixed them — the validation loop working as intended. None of this is a state-of-the-art or official-benchmark claim (Section 6, items 2 and 6). The conflict-resolution result still suggests that an explicit extract-link-consolidate memory architecture addresses a failure mode most retrieval systems share, and the clear TTL gap identifies where nous should improve next; both justify a full-benchmark evaluation. The post-baseline program (Section 5.4) closes the loop the baseline opened: the same replay infrastructure that priced the baseline localized three write-path store defects and a retrieval-granularity defect, drove their upstream fixes (a bounded iterative keyed-retrieval leg; an exemplar-granularity retrieval mode), and verified the final configurations at **CR 0.812** (single-hop 0.938, multi-hop 0.688; p=0.0008) and **ICL 0.695** (p=0.0018; the baseline's sole decidable field loss narrowed from −28.5pp to −14.5pp, with the remaining gap attributed to answer-time reader discretion, not retrieval) — evidence that the benchmark is functioning here as an engineering instrument, not a leaderboard entry. The harness, vendored graders, pinned configuration, and per-question result records are available in the project repository; `BASELINE_SUMMARY.md` reconciles every number to its evidence file and comparability tier.

A Reconstructed judge prompt (verbatim)

Infini Memory's judge prompt is unpublished; its protocol is described only as "a binary correctness judgment based on the question, reference answer, and model output" with gpt-5. Our reconstruction, applied verbatim to every persisted answer (model `gpt-5`, default temperature, no few-shot):

```
You are evaluating whether a model's output correctly answers a question, given the
reference answer.

Question: {question}

Reference Answer: {reference}

Model Output: {output}

Does the model output contain the correct answer according to the reference? The
output may be verbose or phrased differently; judge semantic correctness only.
Respond with exactly one word: yes or no.
```

Multiple reference aliases are joined with " | "; outputs are truncated at 4,000 characters; correctness = the reply begins with "yes". Validation evidence (judge-vs-official agreement, Cohen's kappa, and a second-judge sensitivity run with gpt-4o under the identical prompt) is reported in the repository's `BASELINE_SUMMARY.md` alongside the per-question judge verdict files (`reports/judge_regrade/`).

References

[1] Y. Hu et al., "Evaluating Memory in LLM Agents via Incremental Multi-Turn Interactions" (MemoryAgentBench), ICLR 2026. `github.com/HUST-AI-HYZ/MemoryAgentBench`, arXiv:2507.05257.

[2] "Infini Memory: Maintainable Topic Documents for Long-Term LLM Agent Memory," arXiv:2606.10677, 2026.